ACADEMIC PROJECT REPORT

INVENTORY DEMAND FORECASTING

Inventory Demand Analysis and Forecasting Using Python

Python | Pandas | NumPy | Matplotlib | Streamlit

SUBMITTED BY

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A data analytics project for examining retail demand, product inventory, category and regional performance, and the quality of supplied demand forecasts.

2. Declaration and Acknowledgement

Declaration

We submit this report as documentation of the Retail Inventory Analytics project. The report describes the data-processing pipeline, the analysis implemented in the project scripts, and the visual outputs included in the repository. Quantitative values are calculated from the selected project CSV when the report is generated.

The project evaluates the Demand Forecast field already present in the dataset. It does not claim that a new forecasting model was trained by this report generator. Inventory classifications use the thresholds implemented in the project's analysis code.

Acknowledgement

We thank our faculty and institution for the opportunity to complete this project. We acknowledge the Python open-source ecosystem, particularly Pandas, NumPy, Matplotlib and ReportLab, for supporting data processing, visualization and report creation.

We also acknowledge the project repository's README, cleaning and analysis scripts, cleaned dataset, and existing visualization files as the source material for this report.

Section takeaway
These statements clarify what the report covers and distinguish the work completed in the repository from work that has not been claimed. The final PDF should be checked against the submitted code and data before academic submission. Clear documentation also supports fair review: readers can distinguish implemented features from recommendations for later work, and can reproduce the calculations by following the repository instructions and using the same input files.

3. Abstract

Retailers need to balance product availability against the cost of holding inventory. This project analyzes historical retail inventory and sales records to examine product demand, category and regional revenue, seasonality labels, daily demand trends, and the difference between supplied forecasts and observed units sold.

The repository uses Python and Pandas for data preparation and aggregation, NumPy for numerical operations, Matplotlib and Seaborn for visualizations, and Streamlit for a simple dashboard. Its cleaning stage removes exact duplicate rows, standardizes selected categorical fields, parses dates, handles invalid numeric values, and fills selected missing values according to documented project rules.

The analysis computes revenue as Units Sold × Price × (1 – Discount / 100), summarizes demand by product, applies heuristic stock-status thresholds, and calculates mean absolute error (MAE) between supplied Demand Forecast values and actual Units Sold. These results support exploratory business analysis, but should be interpreted with the dataset's limitations and assumptions in mind.

Report Contents

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4–6	Introduction, problem, objectives and project scope
7–10	Dataset, cleaning, workflow and key metrics
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19–21	Inventory status, findings, recommendations and limitations
22	Conclusion, future scope and references

Section takeaway
In short, the project turns a large spreadsheet into information that is easier to inspect. Its value lies in making patterns visible and giving a reviewer a clear path from the source data to the reported results. The report therefore focuses on transparent methods and interpretable outputs rather than claiming a production-ready forecasting system. Its sections move from source data and preparation to metrics, visual evidence, practical implications and areas that need further validation.

4. Introduction

Inventory management is a core retail activity. A business needs enough stock to satisfy customer demand, but excess stock can increase holding costs and tie up working capital. Products may also differ in their sales volume, price, seasonality and regional demand.

Data analytics helps convert historical records into summaries and visual evidence. Product-level demand analysis can highlight fast-moving items; category analysis can compare sales contribution; time-series views can show whether daily sales fluctuate; and forecast comparisons can indicate how close supplied estimates are to observed sales.

Project Overview

The project, Retail Inventory Analytics, processes a retail inventory workbook covering 2025–2026. Its pipeline separates data cleaning from analysis, saves a cleaned CSV, calculates business metrics and creates charts. A Streamlit dashboard presents key metrics, stock-status counts, charts and product-demand tables.

Business Scenario

Imagine a retail manager opening a large spreadsheet and trying to work out which products sell quickly, which categories contribute to sales, and whether inventory looks unusually low or high. This project organizes those records into summaries and charts, making it easier to spot patterns and decide what deserves a closer look. The results are exploratory indicators, not automatic business decisions.

Project Folder Structure

The repository is arranged so that source data, processing code, generated outputs and the dashboard are easy to locate. The raw workbook is kept separate from the cleaned CSV, while reusable cleaning and analysis logic lives in the src folder. This separation makes the workflow easier to maintain and helps reviewers trace each result back to its source.

Inventory_Forecast-main/
■■■ data/
■ ■■■ rawdata/ — original Excel workbook
■ ■■■ cleandata/ — cleaned CSV dataset
■■■ src/
■ ■■■ data_cleaning.py — cleaning rules
■ ■■■ analysis.py — metrics and charts
■■■ outputs/ — charts and exported data
■■■ main.py — pipeline entry point
■■■ dashboard.py — Streamlit interface
■■■ generate_report.py — Reportlab
■■■ Inventory_Demand_Forecasting.pdf
■■■ requirements.txt — Python dependencies
■■■ README.md — setup and workflow notes

Section takeaway
The main benefit of this workflow is consistency: the same cleaned file and calculation rules can be used again when the data is refreshed. The charts are intended to support human review, not replace business judgement. The folder layout mirrors these responsibilities, reducing the chance that original data is confused with processed output. It also makes it easier for another student or reviewer to identify the entry point, dependencies and scripts without searching through unrelated files.

5. Problem Statement and Objectives

Problem Statement

Raw sales and inventory records can be difficult to interpret without cleaning, aggregation and visualization. A retailer may struggle to identify high-demand products, compare category and regional revenue, recognize broad demand trends, and evaluate the supplied forecast values. This project organizes the available data to make these questions easier to investigate.

Main Objective

To analyze retail inventory and sales data with Python, produce meaningful visualizations, compare supplied demand forecasts with actual units sold, and summarize possible inventory risks using the project's stated rules.

Specific Objectives

- Inspect and prepare the source workbook for analysis.
- Remove exact duplicate rows and handle selected data-quality issues.
- Calculate revenue, unit sales and average price metrics.
- Identify products with the highest and lowest recorded demand.
- Compare revenue and units sold across categories and regions.
- Visualize daily demand with a seven-day moving average.
- Calculate forecast MAE from supplied forecast and actual values.
- Classify products using the project's inventory thresholds.

Section takeaway
These objectives connect the technical tasks to practical questions a retailer may ask. Each objective can be reviewed by checking whether the project produces the relevant table, metric, chart or documented rule. A clear objective list is important because exploratory projects can easily expand beyond their available data. The present scope stays close to the columns and operations implemented in the repository, while more advanced optimization is reserved for future development.

6. Scope, Business Questions and Technology

Business Questions

- Which product IDs have the highest and lowest recorded demand?
- Which product categories contribute most to revenue and units sold?
- How does daily unit sales volume change over time?
- How are units sold distributed across seasonality labels?
- How closely do supplied demand forecasts match actual units sold?
- Which products meet the project's under-stock or over-stock rules?
- How do revenue and category distribution vary by region?

Technology Stack

Technology	Use in this project
Python	Core programming language for the pipeline and dashboard.
Pandas	Data loading, cleaning, grouping and summary calculations.
NumPy	Numerical operations, including forecast-error calculation.
Matplotlib / Seaborn	Generation of static charts and comparisons.
Streamlit	Interactive dashboard interface.
ReportLab	Automated PDF report layout and export.
Excel / CSV	Source workbook and processed data storage.

Scope Boundary

This is an exploratory analytics project, not a real-time inventory-control system. It evaluates the forecasts provided in the dataset; the existing analysis script does not train a forecasting model. Business thresholds should be validated before operational use.

Section takeaway
The questions define the boundary of the current work. They focus on description and comparison; decisions such as how much to order would require additional information about lead times, supplier constraints and service targets. These questions are useful because they connect each chart or summary to a decision-relevant topic. They do not, on their own, establish causation; any proposed action should be checked against stock availability, supplier lead times, pricing strategy and business priorities.

7. Dataset Description and Data Dictionary

The report reads the cleaned CSV from the project-relative path data/cleandata/retail_store_inventory_cleaned.csv when available. The selected CSV contains 73,501 rows and 16 columns at report-generation time. Its date range after parsing is shown below.

Dataset measure	Observed value
Selected file	data\cleandata\retail_store_inventory_cleaned.csv
Records	73,501
Columns	16
Date range	01 Jan 2025 to 01 Jan 2027
Distinct product IDs	20
Distinct categories	6
Distinct regions	4
Missing cells in selected CSV	205
Duplicate rows in selected CSV	195

First Six Rows of the Raw Dataset

The following preview shows the first six records from the original raw Excel workbook. The columns are split into two compact tables so that all source fields remain visible.

Date	Store ID	Product ID	Category	Region	Inventory Level	Units Sold	Units Ordered
2025-01-01	S001	P0001	Groceries	North	231	127.0	55
2025-01-01	S001	P0002	Toys	South	204	150.0	66
2025-01-01	S001	P0003	Toys	West	102	65.0	51
2025-01-01	S001	P0004	Toys	North	469	61.0	164
2025-01-01	S001	P0005	Electronics	East	166	14.0	135
2025-01-01	S001	P0006	Groceries	South	138	128.0	102

Demand Forecast	Price	Discount	Weather Condition	Holiday/Promotion	Competitor Pricing	Seasonality
135.47	33.5	20	Rainy	0	29.69	Autumn
144.04	63.01	20	Sunny	0	66.16	Autumn
74.02	27.99	10	Sunny	1	31.32	Summer
62.18	32.72	10	Cloudy	1	34.74	Autumn
9.26	73.64	0	Sunny	0	68.95	Summer
139.82	76.83	10	Sunny	1	79.35	Winter

Main Fields

Field	Description based on project usage
Date	Sales or inventory date; used for daily trend analysis.
Store ID	Store identifier in the source dataset.
Product ID	Product identifier used for product-level summaries.
Category	Product category used for revenue and demand grouping.
Region	Region label used for regional comparisons.
Inventory Level	Recorded inventory measure used for stock-status proxy.
Units Sold	Observed sales volume / demand measure.
Demand Forecast	Forecast values supplied in the dataset.
Price / Discount	Inputs to the project's revenue formula.
Seasonality	Label used to group units sold by seasonality.

Section takeaway
The raw sample is included to make the source structure transparent. The cleaned dataset may differ from these original rows because the cleaning stage standardizes fields and handles selected invalid or missing values. The sample should be treated as a small illustration, not as a representative summary of every product or period. The full dataset is used for the calculations, and the displayed rows help readers understand the original column names and value formats.

8. Data Cleaning and Preprocessing

The cleaning rules below are implemented in src/data_cleaning.py and described in the repository README. They are project-specific rules rather than universal inventory policies.

Issue	Implemented handling
Exact duplicate records	Drop exact duplicate rows.
Category, Region, Weather Condition, Seasonality	Trim whitespace, title-case labels, and convert text representations of missing values to null.
Invalid dates	Parse dates with coercion and drop rows whose dates cannot be parsed.
Negative Inventory Level	Treat as missing, then fill with the dataset median.
Negative Units Sold	Treat as missing, then fill with the dataset median.
Discount outside 0–100	Treat as missing; the script does not impute this field.
Price above $1,000	Treat as missing, then fill using the category median.
Missing Category	Fill with 'Uncategorized'.

Data Quality Notes

The report's missing-cell and duplicate counts describe the CSV that is actually read when the report runs. They do not represent the raw workbook unless that raw workbook is selected. The cleaning script does not currently impute every missing field, so some null values may remain in the cleaned CSV.

Why Cleaning Matters

Duplicate rows can inflate sales totals, invalid dates can distort time trends, and invalid numeric values can make demand or inventory comparisons unreliable. Cleaning decisions should be recorded because imputation can affect downstream totals.

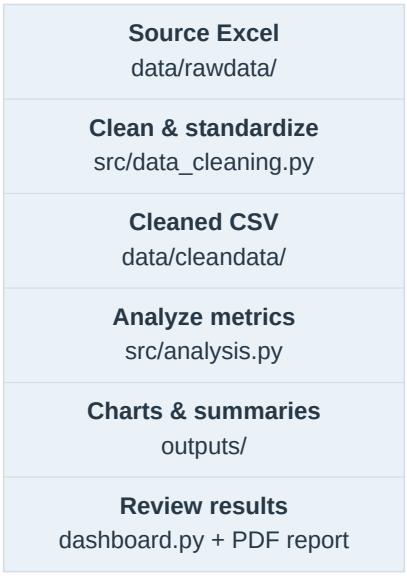
> **Section takeaway**
> Cleaning improves consistency, but it does not make every record perfect. Median filling is a practical choice for this project and can influence averages and totals, so the cleaning rules should remain visible to anyone interpreting the findings. The cleaned output should still be reviewed for unusual values and remaining missing fields. In a business deployment, cleaning rules would ideally be documented alongside row counts before and after each step so that data changes can be audited.

9. Project Workflow and Methodology

Stage	Project action	Purpose
1. Load	Read source Excel workbook.	Make records available for processing.
2. Clean	Remove duplicates; standardize labels; parse dates; handle selected invalid values.	Improve consistency.
3. Export	Save cleaned records as CSV.	Create reusable analysis input.
4. Calculate	Compute revenue, product summaries, stock status and forecast MAE.	Answer business questions.
5. Visualize	Save charts to outputs/.	Communicate comparisons and trends.
6. Dashboard	Display metrics, charts and product summaries in Streamlit.	Provide a user-facing view.
7. Report	Summarize project design, data and results in PDF.	Document findings and limitations.

End-to-End Process Flow

The README identifies main.py as the recommended entry point. It coordinates the cleaning stage, writes the cleaned dataset, runs the analysis, and refreshes the saved visual outputs. The dashboard then presents selected results for exploration, while this PDF documents the process, interpretation and limitations. Each stage can also be inspected separately when debugging or validating an output.



Revenue Calculation

$$\text{Total Revenue} = \text{Units Sold} \times \text{Price} \times (1 - \text{Discount} / 100)$$

The formula applies the discount percentage to the row-level price and multiplies by units sold. The repository notes that this calculation does not account for taxes, returns, shipping or other accounting adjustments.

Forecast Evaluation

$$\text{MAE} = \text{Mean of (Actual Units Sold} - \text{Demand Forecast)}$$

The MAE measures average absolute difference in units. It evaluates the forecast values already present in the dataset; it does not prove that the forecast is unbiased or accurate for every product, region or season.

Section takeaway
Keeping the stages separate makes the project easier to debug and maintain. If an output looks unexpected, the data, cleaning rules, aggregation formula and chart code can be inspected independently. This staged design supports reproducibility: the same entry point can be run again after replacing the input workbook, and the derived outputs can be refreshed. The README also documents separate commands for running cleaning or analysis independently when troubleshooting.

10. Exploratory Data Analysis and Key Metrics

Exploratory Data Analysis summarizes the data before interpreting individual charts. The values below are calculated dynamically from the CSV selected by this script.

Metric	Observed value
Total units sold	10,028,853
Total revenue (dataset currency)	497,274,414.33
Average row-level unit price	55.13
Forecast MAE (units)	8.39
Rows with actual and forecast values	73,455
Product IDs	20
Categories	6
Regions	4

How to Read These Metrics

Total units sold summarizes recorded volume. Total revenue follows the project's discount formula. Average unit price is an unweighted mean of row-level prices, not a sales-weighted average. MAE is an aggregate error measure and should be interpreted alongside product-level errors and the scale of demand.

Data Caveat

The figures depend on the selected input file and its current contents. If the dataset or cleaning rules change, rerun the report to refresh these results.

Section takeaway
Summary metrics are most useful when their definitions are clear. They describe the dataset that was actually loaded at report-generation time, so values can change if the input file is updated. Counts and averages should be read together because no single metric captures the full business picture. Revenue can be influenced by both volume and price, while forecast error should be compared with typical demand levels and a simple baseline.

11. Revenue by Product Category

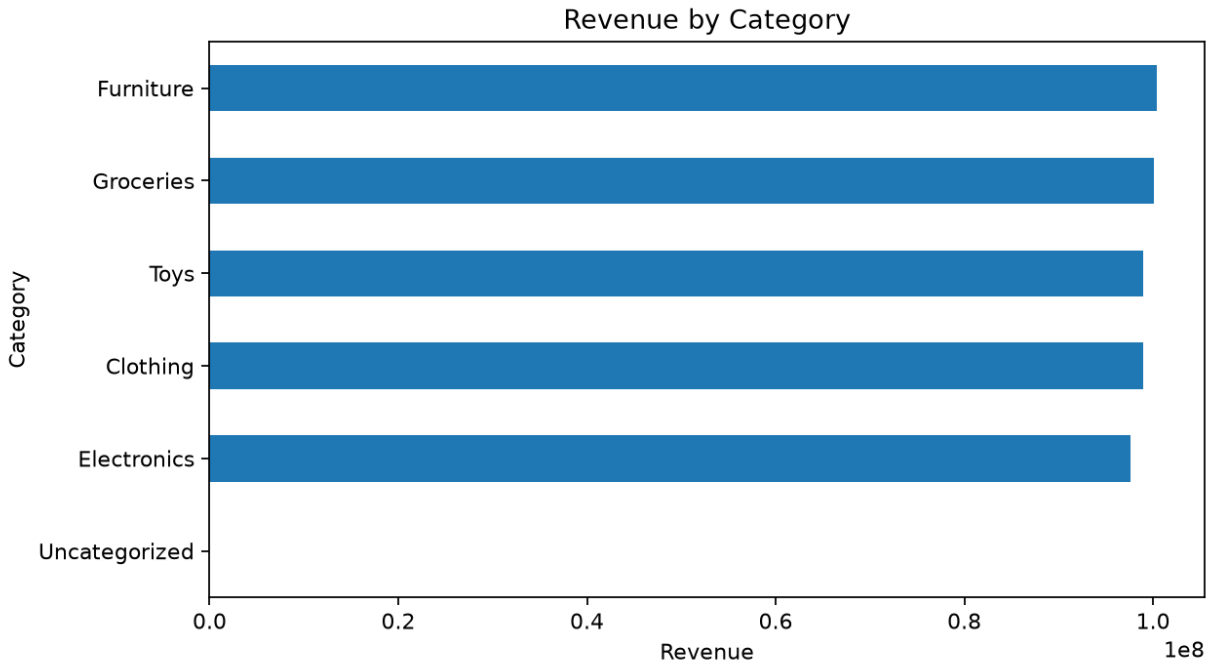


Figure 1. Total revenue distribution by product category. The original chart is included from outputs/.

Category Revenue Summary

Category	Total Revenue
Furniture	100,705,529.37
Groceries	100,352,276.82
Clothing	99,284,548.41
Toys	99,134,798.89
Electronics	97,797,260.83
Uncategorized	0.00

Category revenue helps identify which product groups contribute the most recorded sales value. Management can review the strongest categories and investigate lower-contributing categories. Revenue alone does not indicate profit because product costs and operating expenses are not included in this project metric.

Section takeaway
A chart is a starting point for investigation rather than proof of a cause. The pattern should be compared with the underlying data and interpreted alongside the business context before drawing conclusions. Category totals may be affected by differences in product prices, discount rates and the number of records in each group. A useful follow-up would compare both units and revenue and examine whether the pattern remains consistent across dates and regions.

12. Units Sold by Seasonality

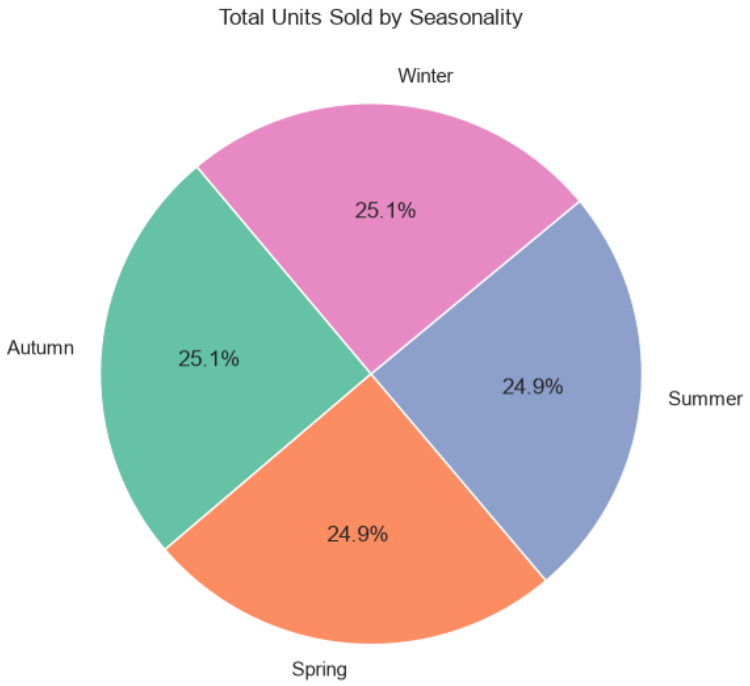


Figure 2. Distribution of recorded units sold across the Seasonality labels in the dataset.

The seasonality chart groups units sold by the Seasonality field. It is a descriptive comparison of the labels present in the dataset. It should not automatically be interpreted as proof of yearly seasonal behavior, because that conclusion requires repeated observations across enough time periods and a clear definition of each season label.

Business Use

If demand differences are consistent and supported by historical observations, a retailer can review purchasing plans before expected high-demand periods. The chart should be considered alongside date-level demand, promotions and stock availability.

Section takeaway
Seasonality labels provide a simple way to compare groups in the source data. They should not be confused with a formal seasonal forecasting model, which would require a defined time-series method and evaluation process. Seasonal groupings are descriptive labels inherited from the source data. They can help compare observed sales across groups, but a stronger seasonal conclusion would require checking the time coverage, label definitions and repeated patterns across multiple years.

13. Highest-Demand Products

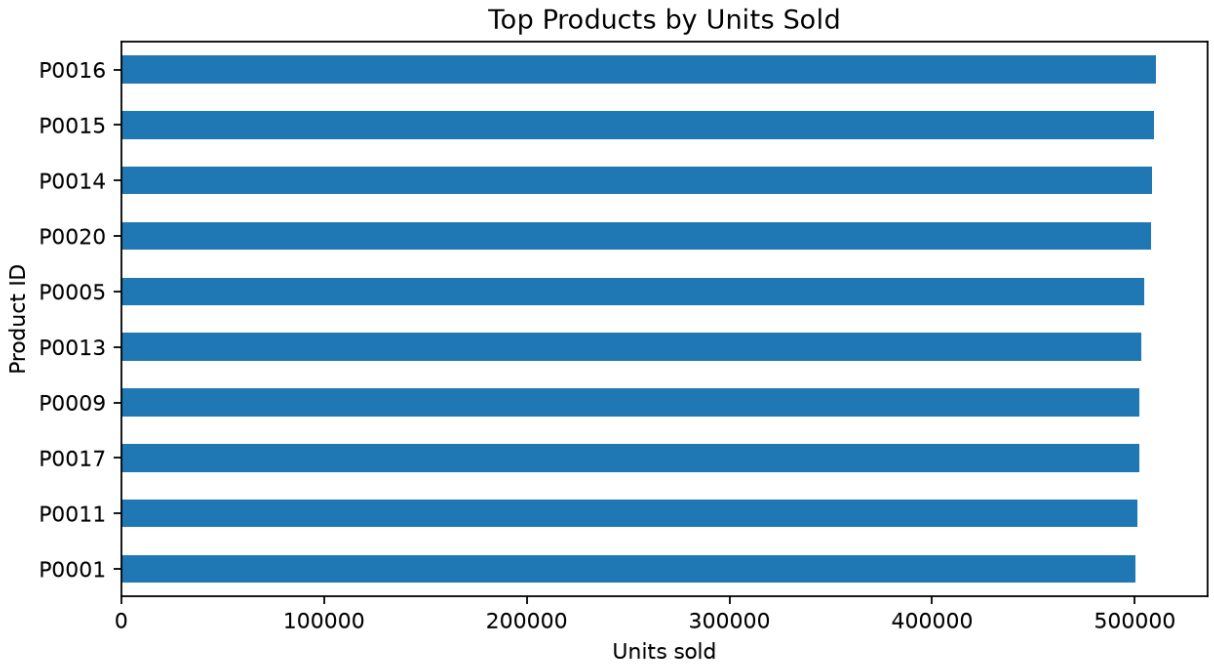


Figure 3. The ten product IDs with the highest total recorded demand.

Top Products by Total Demand

Product ID	Total Demand	Average / Row	Avg. Inventory	Avg. Forecast
P0016	511,891	138.95	277.29	143.84
P0015	511,377	138.96	276.41	144.04
P0014	510,122	138.85	278.00	143.84
P0020	509,998	138.93	276.79	143.64
P0005	506,490	137.78	272.97	142.82
P0013	504,228	137.13	275.25	142.46
P0009	503,376	137.20	275.15	142.39
P0017	503,372	136.93	278.35	141.99

High total demand can indicate products that deserve frequent stock monitoring. The average demand and current inventory proxy add context, but replenishment decisions should also consider supplier lead time, order constraints and stockout history.

Section takeaway
Product-level rankings help narrow attention to items with unusually high or low recorded demand. Reviewing the underlying time periods and inventory availability can help explain whether the ranking reflects a stable pattern or a temporary spike. High-demand products deserve attention, but a ranking does not show whether stock was available whenever customers wanted to buy. Reviewing stockout periods and the dates behind each product total would help distinguish persistent demand from isolated peaks.

14. Daily Demand and 7-Day Moving Average

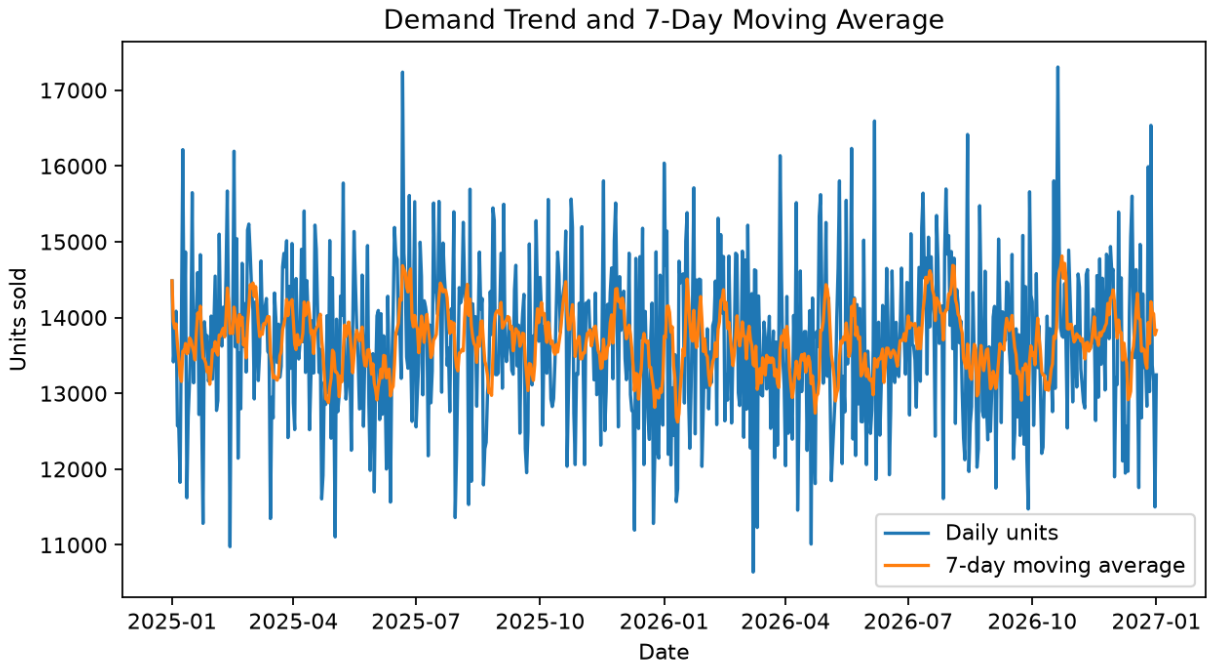


Figure 4. Daily total units sold with the seven-day moving-average line.

Moving Average

7-day moving average = mean of the most recent seven daily observations

The moving average smooths some short-term variation so that broader changes are easier to see. It can help identify sustained rises or falls in demand. It can also respond slowly to sudden changes and is meaningful only when the observations are correctly ordered and represent consistent daily intervals.

Interpretation Guidance

Read the daily series and smoothed line together. Large gaps or spikes should be investigated rather than automatically treated as normal demand behavior.

Section takeaway
The moving average smooths daily variation and makes the broad direction easier to see. It can also hide short-lived peaks, so the original daily series remains important when planning for sudden changes. Moving averages are useful for communication because they reduce day-to-day noise, although the chosen window affects how quickly the line responds to change. Alternative windows could be compared to decide which is most informative for the intended planning horizon.

15. Supplied Forecast Versus Actual Units Sold

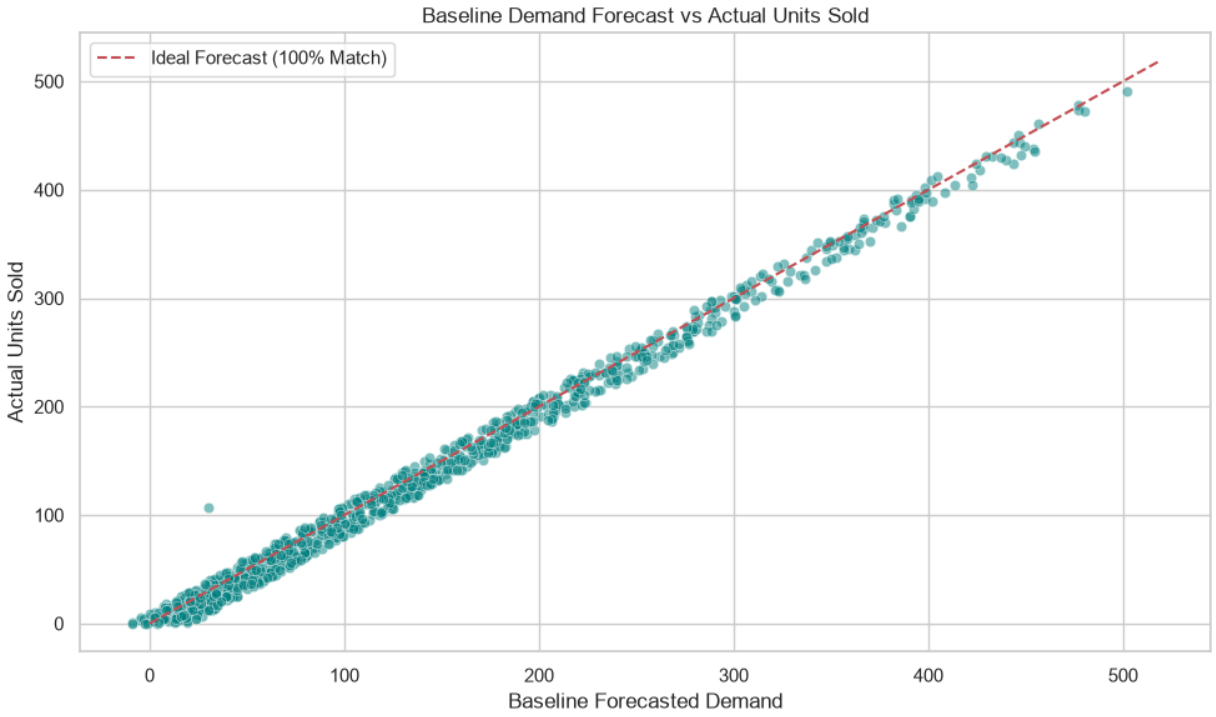


Figure 5. Scatter plot comparing supplied Demand Forecast values with observed Units Sold.

Evaluation metric	Value from selected CSV
Mean Absolute Error (MAE)	8.39 units
Rows with both values available	73,455

Interpretation

In the scatter plot, points close to the ideal equality line represent rows where the forecast and actual units are similar. Points far from the line indicate larger differences. MAE summarizes the average absolute error, but does not show whether the model tends to overpredict or underpredict.

The existing project evaluates forecasts supplied in the dataset. It does not train a new forecasting model. Stronger evaluation could include MAE by product and period, RMSE, bias, and a time-based holdout test.

Section takeaway
MAE is easy to explain because it reports an average absolute difference in units. It does not show the direction of errors, and it should ideally be compared with a baseline before judging forecast quality. Forecast evaluation would be stronger if the project documented how the supplied forecasts were created and compared them with a naïve baseline. Error should also be examined by product and date to avoid hiding weak performance within an overall average.

16. High-Demand Product Categories

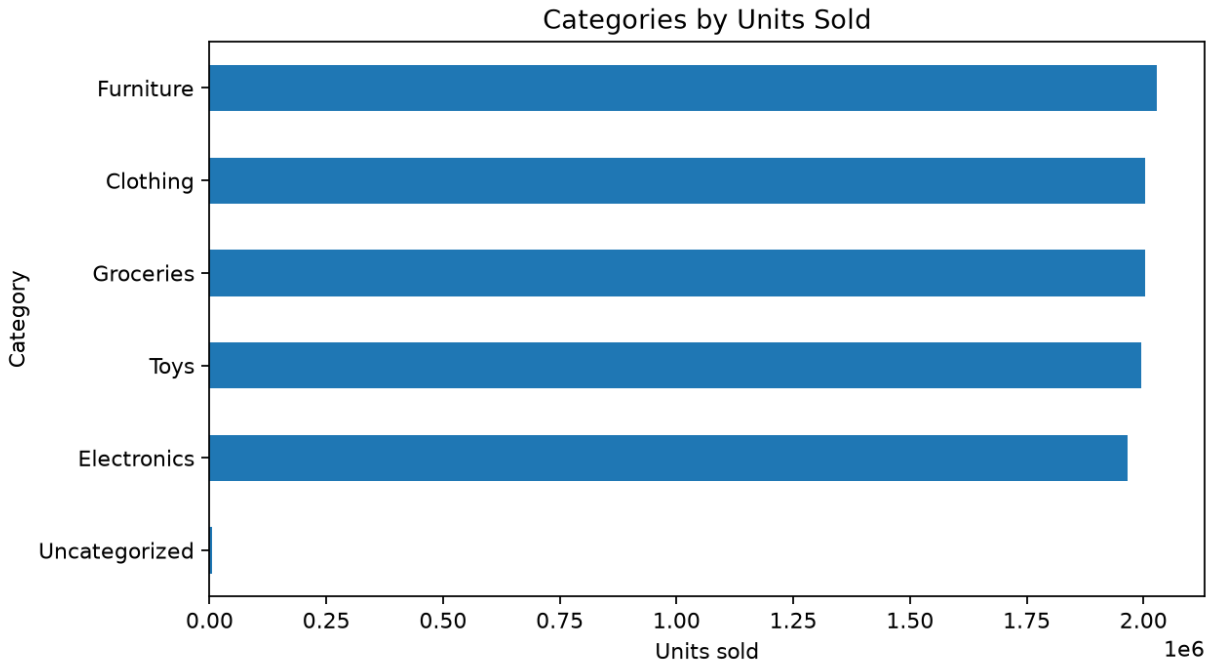


Figure 6. Product categories ranked by total recorded units sold.

Units Sold by Category

Category	Total Units Sold
Furniture	2,035,288
Clothing	2,010,640
Groceries	2,009,373
Toys	1,999,562
Electronics	1,969,461
Uncategorized	4,529

This view ranks categories by sales volume rather than revenue. A category can sell many units without being the largest revenue contributor if its prices are lower. Comparing this chart with category revenue gives a more complete picture of sales performance.

Section takeaway
Unit volume and revenue answer different questions. A category with many low-priced sales may lead in units while another category contributes more revenue, so both views should be considered together. Considering unit volume alongside revenue gives a more balanced view of category performance. The results can guide further investigation, but decisions about assortment or promotion should also consider margins, customer needs, stock availability and operating costs.

17. Regional Revenue Analysis

Chart not found: outputs/07_region_revenue.png

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Regional Revenue Summary

Region	Total Revenue
East	125,431,218.48
South	125,178,247.29
North	123,977,794.72
West	122,687,153.84

Regional comparison can highlight differences in recorded sales contribution. Before making regional decisions, consider the number of stores, product mix, promotions, demand coverage and inventory availability represented in each region.

Section takeaway
Regional differences can reflect product mix, store coverage, promotions or data availability. A fair comparison should consider these factors before interpreting a region as stronger or weaker. Regional comparisons are most informative when the underlying coverage is comparable. If regions contain different numbers of stores or records, totals may need to be supplemented with per-store or per-period measures before drawing conclusions about relative performance.

18. Regional and Category Distribution

Chart not found: outputs/08_region_category_countplot.png

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This visualization provides a combined regional and category view. It can help identify whether the mix of recorded products or observations differs by region. The exact interpretation should follow the axes, legend and counting method shown in the original chart.

Potential Business Use

If category demand varies by region, the retailer can investigate whether local assortment or replenishment plans should differ. Any change should be based on demand and stock measures, not chart appearance alone.

> **Section takeaway**
> The combined view can reveal differences in category mix across regions. It is useful for deciding where to investigate further, but the underlying counts and chart labels should be checked before making assortment changes. A combined region-category view helps identify where category mixes differ, which can support targeted questions for managers. The chart should be used to prioritize investigation rather than as a stand-alone reason to change stock allocations.

19. Product-Level Inventory Status

Stock status	Product count	Rule implemented in project
Under-stocked	20	Mean inventory < 3 × average demand
Optimal	0	Does not meet either threshold
Over-stocked	0	Mean inventory > 15 × average demand

Product Summary — Highest Demand

Product ID	Total Demand	Avg Demand	Mean Inventory	Status
P0016	511,891	138.95	277.29	Under-stocked
P0015	511,377	138.96	276.41	Under-stocked
P0014	510,122	138.85	278.00	Under-stocked
P0020	509,998	138.93	276.79	Under-stocked
P0005	506,490	137.78	272.97	Under-stocked
P0013	504,228	137.13	275.25	Under-stocked
P0009	503,376	137.20	275.15	Under-stocked
P0017	503,372	136.93	278.35	Under-stocked
P0011	502,581	136.68	272.97	Under-stocked
P0001	501,049	136.23	272.74	Under-stocked

These classifications follow the repository's heuristic: average inventory per product is compared with three and fifteen times that product's average recorded demand. They are not formal replenishment rules. Real inventory policies should account for lead time, safety stock, service level, stockouts and replenishment frequency.

Dashboard Visualization



Figure 9. Dashboard overview.

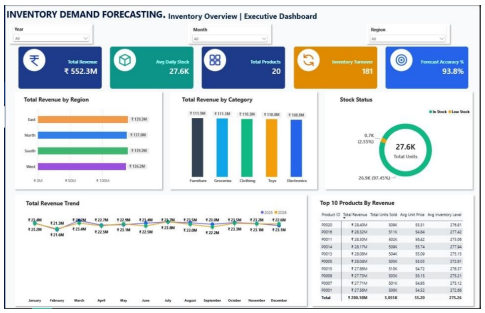


Figure 10. Additional dashboard visualization.

Section takeaway
The stock labels are screening signals created by simple thresholds. They are useful for identifying records to review, but should not be treated as purchase instructions without operational context. Thresholds should be reviewed with operational stakeholders before use in replenishment. Real stock decisions commonly depend on supplier lead time, minimum order quantity, demand variability, service targets and current purchase orders, none of which is fully represented by this simple heuristic.

20. Findings and Business Insights

Findings from the Implemented Analysis

- The selected data file contains 73,501 records and 16 columns.
- The selected CSV has 205 missing cells and 195 exact duplicate rows at report time.
- The dataset includes 20 distinct product IDs across 6 categories and 4 regions.
- Total recorded units sold are 10,028,853.
- Revenue calculated with the project formula is 497,274,414.33 in the dataset's currency units.
- The supplied forecasts have an aggregate MAE of 8.39 units over 73,455 usable rows.
- The project's inventory heuristic classifies 20 products as under-stocked, 0 as optimal, and 0 as over-stocked.

Business Insights

The product and category tables can guide attention toward high-volume products and important categories. The demand trend can support review of recent movement. Forecast error provides a baseline signal of how far supplied forecasts differ from observed sales. Inventory status can be used as a screening indicator for products requiring further review.

These are analytical indicators rather than guaranteed business outcomes. The dataset may not contain all information needed to explain demand changes or optimize purchasing.

Section takeaway
The findings bring together the main outputs of the implemented analysis. They are descriptive results from this dataset, and their reliability depends on the quality, coverage and assumptions of the source records. The findings are strongest when they are traceable to a visible metric, table or chart. Any final recommendation should state the evidence behind it, acknowledge data limitations and specify what additional information would be needed to make the decision more reliable.

21. Recommendations, Limitations and Future Scope

Recommendations

- Review the highest-demand products regularly and compare demand with current stock.
- Use regional and category summaries to identify where deeper investigation is needed.
- Validate missing values and duplicate rows before relying on totals.
- Evaluate forecast errors separately by product and time period.
- Compare simple moving-average baselines with other forecasting methods.
- Calibrate stock-status thresholds against supplier lead times and desired service levels.
- Keep the source workbook, cleaned CSV, scripts and generated charts version-aligned.

Limitations

- Historical demand may not represent future customer behavior.
- Median imputation can affect sales and inventory summaries.
- The project uses simple inventory thresholds that are not calibrated operational policies.
- Aggregate MAE does not reveal forecast bias or performance for each product.
- Revenue excludes taxes, returns, shipping and other accounting adjustments.
- External factors may affect demand but are not fully modeled in the current pipeline.

Future Scope

- Compare 7-, 14- and 30-day moving-average forecasts.
- Add weighted moving average or simple exponential smoothing.
- Report MAE, RMSE and forecast bias by product and period.
- Estimate safety stock and reorder points using lead time and service level.
- Add category-specific forecasts and interactive dashboard filters.

Section takeaway
A sensible next step is to validate the most important findings with domain knowledge, then improve the pipeline incrementally. Better evaluation and clearer stock policies would make the project more useful for real decisions. Future improvements should be introduced in small, testable steps, with each change evaluated against the current implementation. Keeping a documented baseline will make it easier to determine whether a new forecast, metric or inventory rule actually improves the quality of decisions.

22. Conclusion and References

Conclusion

The Retail Inventory Analytics project demonstrates a practical Python workflow for cleaning retail records, summarizing product demand, calculating revenue, reviewing inventory indicators, comparing supplied forecasts with observed sales, and communicating patterns through charts and a Streamlit dashboard. The analysis is organized into reproducible steps and can be rerun when the dataset is updated.

The results can support exploratory inventory review, but should not be treated as automatic purchasing instructions. Data quality, forecast limitations and the heuristic nature of stock thresholds must be considered before applying the findings in a real retail operation. A useful next iteration would test alternative forecasting baselines, report errors over time, and tune inventory rules with realistic supplier lead times and service targets. This would build on the current work without overstating what the existing analysis can prove.

References

GitHub Repository

- Github Repo Link :

END OF REPORT